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C07D 471/20; C12N 9/1096

(54) **CHEMICAL PROCESS FOR PREPARING SPIROINDOLONES AND INTERMEDIATES THEREOF**

CHEMISCHES VERFAHREN ZUR HERSTELLUNG VON SPIROINDOLONEN UND ZWISCHENPRODUKTE DAVON

PROCÉDÉ CHIMIQUE POUR LA PRÉPARATION DE SPIROINDOLONES ET INTERMÉDIAIRES CORRESPONDANTS

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(56) References cited:

WO-A1-2008/050207 WO-A1-2009/132921
GB-A- 974 895 US-A- 2 947 758
US-A- 3 282 959 US-A- 3 953 442

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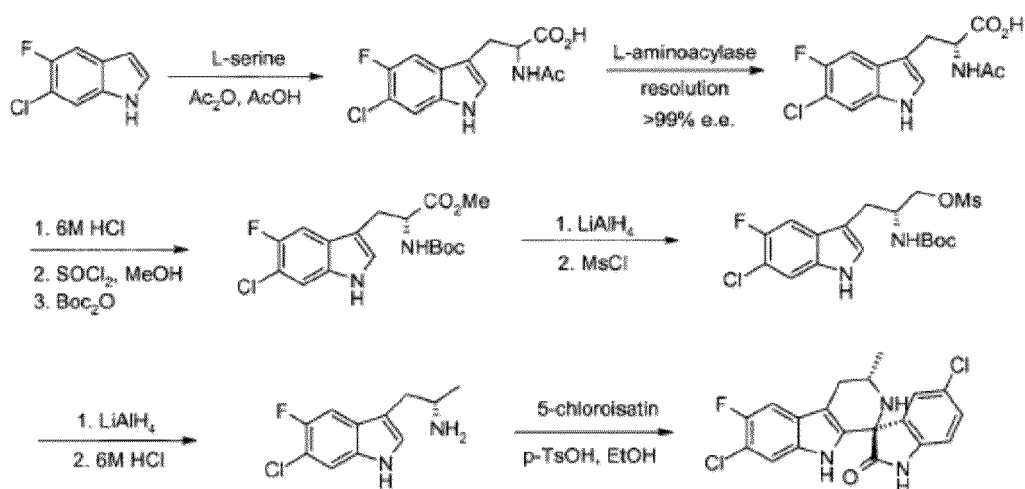
Description

REFERENCE TO SEQUENCE LISTING, TABLE OR COMPUTER PROGRAM

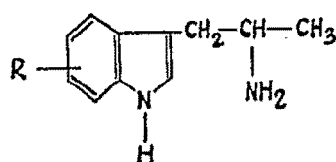
[0001] The official copy of the Sequence Listing is submitted concurrently with the specification as an ASCII formatted text file via EFS-Web, with a file name of "PAT055051_seq12.txt", a creation date of March 22, 2013, and a size of 447 kilobytes. The Sequence Listing filed via EFS-Web is part of the specification.

Prior art:

[0002] (1'R,3'S)-5,7'-dichloro-6'-fluoro-3'-methyl-2',3',4',9'-tetrahydrospiro[indoline-3,1'-pyrido[3,4-b]indol]-2-one (eg. a compound of formula (IV), which comprises a spiroindolone moiety) and a 6-steps synthetic method for preparing, including known chiral amine intermediate compound (IIA) are known (WO 2009/132921):

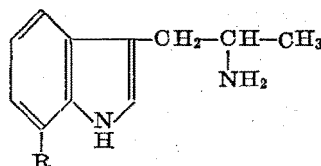


[0003] GB 974 895 A describes a process for production of tryptamine compounds which in free base form have the formula,



and acid addition salts thereof, where R represents a lower alkyl group containing 1 to 3 carbon atoms.

[0004] US 3 282 959 describes tryptamine compounds which in free base form have the formula



where R represents methyl, methoxy or chloro, as well as addition salts thereof; and to processes for production of the same.

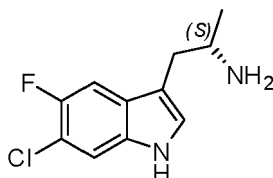
[0005] D. E. Nichols et al., J. Med. Chem., 1988, 31(7), pages 1406-1412 describes a procedure for the preparation of optically pure α -methyltryptamines (AMTs) from substituted indoles.

Invention:

[0006] The invention is defined by the appended claims

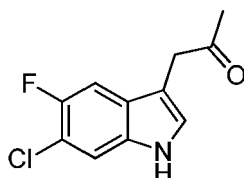
[0007] The present invention is directed to intermediates used in an improved method of synthesizing spiroindolone compounds, in particular, (1'*R*,3'*S*)-5,7'-dichloro-6'-fluoro-3'-methyl-2',3',4',9'-tetrahydrospiro[indoline-3,1'-pyrido[3,4-*b*]indol]-2-one.

[0008] A compound of the invention may be used in a process for preparing a compound of formula (IIA) or a salt or hydrate or solvate thereof,



(IIA)

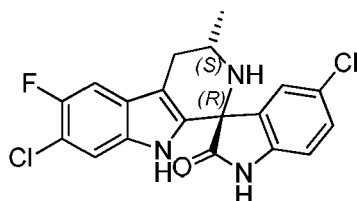
comprising enzymatically transaminating a compound of formula (IA) or a salt or solvate or hydrate thereof,



(IA).

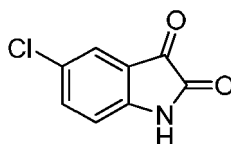
to provide the compound of formula (IIA).

[0009] A compound of the invention may be used in a process for preparing a compound of formula (IVA), or a salt or solvate or hydrate thereof,



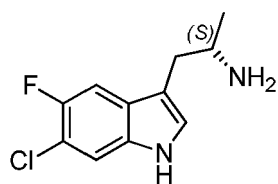
(IVA)

comprising: reacting a compound of formula (IIIA) or a salt or solvate or hydrate thereof,



(IIIA)

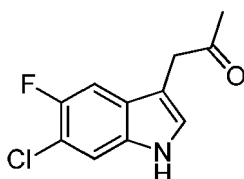
with a compound of formula (II) or a salt or solvate or hydrate thereof,



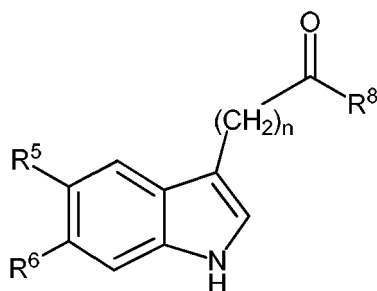
(IIA)

to provide the compound of formula (IVA), or a salt or solvate or hydrate thereof,

wherein the compound of formula (IIA) is prepared from the compound of formula (IA), or a salt or hydrate or solvate thereof,



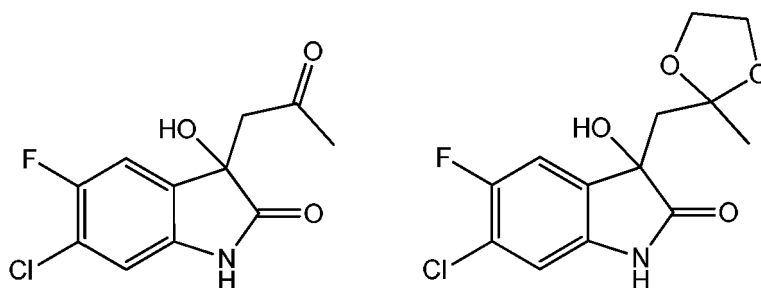
[0010] In an embodiment, the invention is a compound of formula (IC):



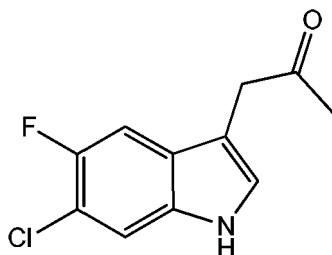
(IC)

wherein: R^5 and R^6 are each, independently halo, hydroxyl, (C_1-C_6) alkyl, trihalo (C_1) alkyl, cyano or (C_1-C_6) alkoxy; R^8 is (C_1-C_6) alkyl, optionally substituted with a hydroxyl; n is 1 or 2; or a pharmaceutically acceptable salt or hydrate or solvate thereof.

[0011] In an embodiment the invention is a compound selected from:



or



or a pharmaceutically acceptable salt, solvate or hydrate thereof.

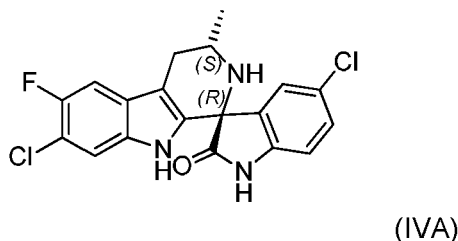
Field of the invention

[0012] The invention relates to a novel intermediate useful for the preparation of spiroindolone compounds useful for the treatment of parasitic diseases comprising e.g. a spiroindolone moiety, such as *(1'R,3'S)*-5,7'-dichloro-6'-fluoro-3'-methyl-2',3',4',9'-tetrahydrospiro[indoline-3,1'-pyrido[3,4-b]indol]-2-one.

Background of the invention

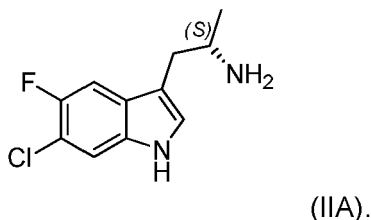
[0013] The present invention is useful in processes for the preparation of spiroindolone compounds, such as *(1'R,3'S)*-5,7'-dichloro-6'-fluoro-3'-methyl-2',3',4',9'-tetrahydrospiro[indoline-3,1'-pyrido[3,4-b]indol]-2-one.

[0014] *(1'R,3'S)*-5,7'-dichloro-6'-fluoro-3'-methyl-2',3',4',9'-tetrahydrospiro[indoline-3,1'-pyrido[3,4-b]indol]-2-one is useful in the treatment and/or prevention of infections such as those caused by *Plasmodium falciparum*, *Plasmodium vivax*, *Plasmodium malariae*, *Plasmodium ovale*, *Trypanosoma cruzi* and parasites of the *Leishmania* genus such as, for example, *Leishmania donovani*., and it has the following structure:



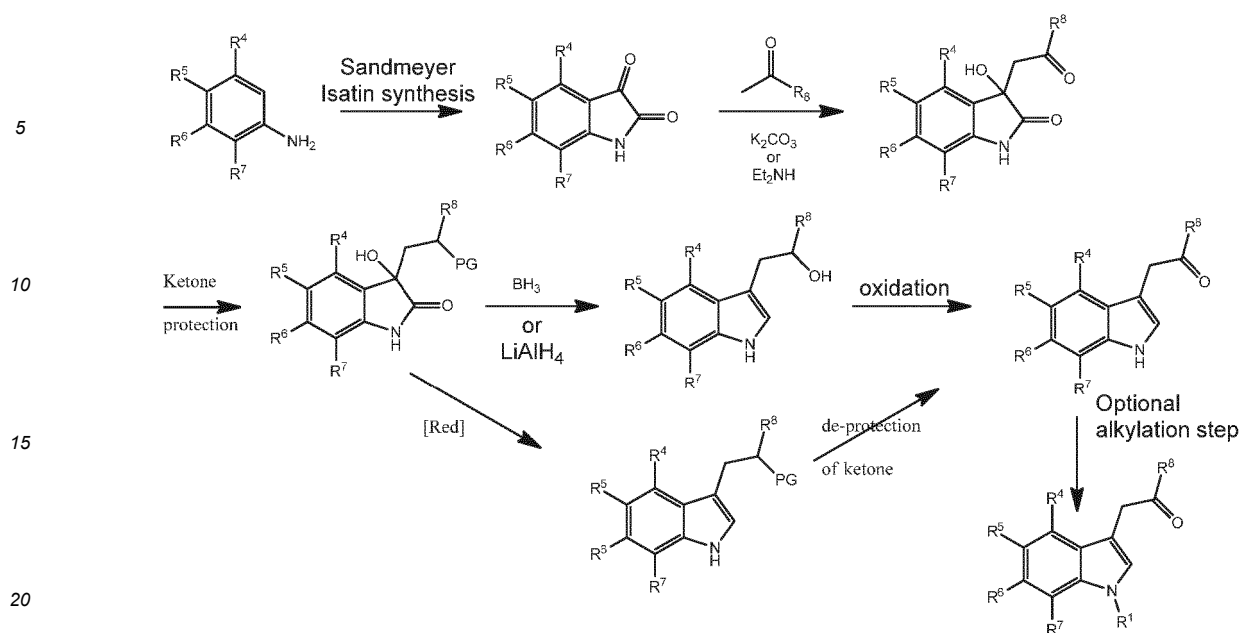
[0015] *(1'R,3'S)*-5,7'-dichloro-6'-fluoro-3'-methyl-2',3',4',9'-tetrahydrospiro[indoline-3,1' pyrido[3,4-b]indol]-2-one and a synthesis thereof are described in WO 2009/132921 A1 in particular in Example 49 therein.

[0016] There is a need to provide new process for the preparation of *(1'R,3'S)*-5,7'-dichloro-6'-fluoro-3'-methyl-2',3',4',9'-tetrahydrospiro[indoline-3,1'-pyrido[3,4-b]indol]-2-one in order to improve the overall efficiency of the synthesis to make it suitable for manufacturing. In particular, there is a need to increase the efficiency of synthesizing the chiral amine intermediate (IIA):



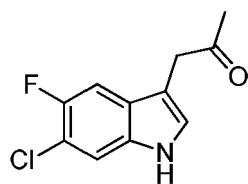
Detailed Description of the Invention

[0017] Process(es) for producing spiroindolone compounds, such as compounds according to formula (IV), or salt or hydrate or solvate thereof, and intermediates, as defined herein, are summarized in Scheme 1.



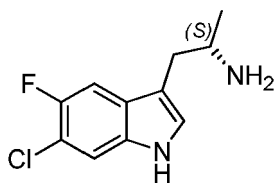
Section B: Conversion of a compound of formula (I) into a compound of formula (II).

[0025] A compound of the invention may be used in a process for converting a compound of formula (I) into a compound of formula (II), wherein, the compound of formula (I) is a compound of formula (IA) or a salt or hydrate or solvate thereof,



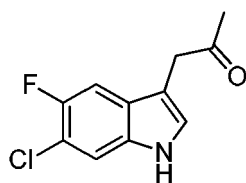
(IA).

[0026] A compound of the invention may be used in a process for preparing a compound of formula (IIA) or a salt or hydrate or solvate thereof,



(IIA)

comprising enzymatically transaminating a compound of formula (IA) or a salt or solvate or hydrate thereof,



(IA).

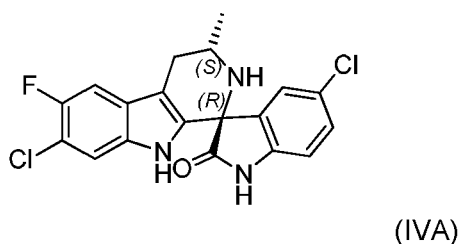
to provide the compound of formula (IIA).

[0027] Typically, the ketone, compound (I) is dissolved in an organic solvent, e.g. glycol, and added to an aqueous mixture of isopropyl amine HCL and pyridoxalphosphate, followed by the addition of TEA buffer. The pH is then adjusted to neutral with an appropriate base, e.g. NaOH followed by warming and addition of the transaminase. The reaction is allowed to stir at temperature for approximately 24 hrs. The solid chiral amine product (compound of formula (II))) is isolated by means known to one of skill in the art, and/or according to examples 10-12.

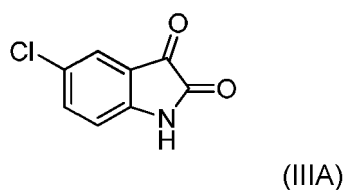
[0028] The enzyme may be SEQ ID NO: 134.

Section C: Conversion of a compound of formula (II) into a compound of formula (IV)

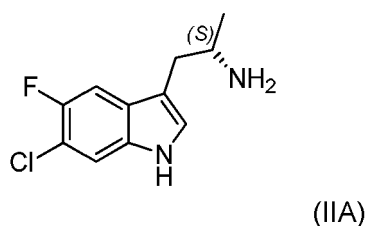
[0029] A compound of the invention may be used in a process for preparing a compound of formula (IVA), or a salt or solvate or hydrate thereof,



comprising: reacting a compound of formula (IIIA) or a salt or solvate or hydrate thereof,

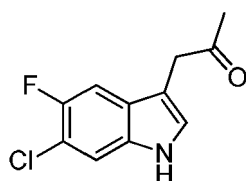


with a compound of formula (II) or a salt or solvate or hydrate thereof,



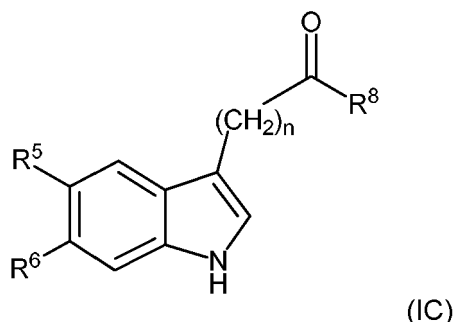
to provide the compound of formula (IVA), or a salt or solvate or hydrate thereof,

wherein the compound of formula (IIA) is prepared from the compound of formula (IA), or a salt or hydrate or solvate thereof,



Section D: Use of the Novel and Inventive Compounds of Formula (IC).

[0030] In an embodiment, the invention is a compound of formula (IC):



wherein: R⁵ and R⁶ are each, independently halo, hydroxyl, (C₁-C₆) alkyl, trihalo (C₁) alkyl, cyano or (C₁-C₆) alkoxy; R⁸ is (C₁-C₆) alkyl, optionally substituted with a hydroxyl; n is 1 or 2; or a pharmaceutically acceptable salt or hydrate or solvate thereof.

[0031] In an alternative embodiment, the invention is a compound of formula (IC) wherein, R⁵ and R⁶ are fluoro when: R⁸ is -CH₃, and n is 1.

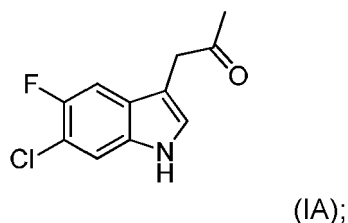
[0032] In an alternative embodiment, the invention is a compound of formula (IC) wherein, R⁵ and R⁶ are fluoro and chloro, when: R⁸ is -CH₃, and n is 1.

[0033] In an alternative embodiment, the invention is a compound of formula (IC) wherein, R⁵ and R⁶ are hydrogen when: R⁸ is -CH₃ and n is 1.

[0034] In an alternative embodiment, the invention is a compound of formula (IC) wherein, R⁵ is fluoro when: n is 1, and R⁶ is hydrogen.

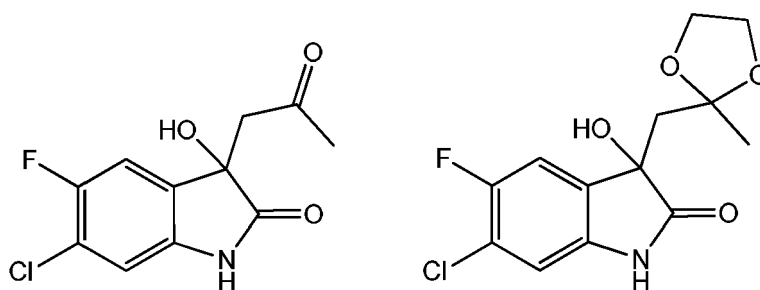
[0035] In an alternative embodiment, the invention is a compound of formula (IC) wherein, R⁵ is H, -OH, -CH₃, -OCH₃, -F, -Cl, -CF₃ or -CN; R⁶ is H, -OH, -OCH₃, -F or -Cl; R⁸ is H, -CH₃, -CH₂CH₃, -CH₂OH, -CO₂H, -CO₂CH₃, -CO₂CH₂CH₃ or -CF₃; and n is 1 or 2.

[0036] In an alternative embodiment, the invention is a compound of formula (I) wherein, the compound is of formula (IA):



or a pharmaceutically acceptable salt or hydrate or solvate thereof.

[0037] In an embodiment the invention is a compound selected from:



or